

Experience

- ML Research Engineer (Contract) — Sherpa, London** 2025–
- Designed and delivered proof-of-concept for LLM integration into survey analysis product.
 - Built custom chain-of-thought pipeline converting natural language queries to JSON requests.
 - Delivered experiments comparing models for feasibility study and strategic recommendations.
- Postgraduate Researcher — University College London** 2024–
- Research on Neural Program Synthesis and Algorithmic Reasoning in the Machine Intelligence Lab, supervised by Prof. Mirco Musolesi. Developed novel deep learning framework.
- Research Intern — Imperial College London** 2023
- Causal discovery using cross-validation as compared with Bayesian Model Selection (using GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP. HPC experience.

Education

- University College London** September 2023 – September 2024
MSc Machine Learning (*Distinction*)
MSc Thesis (*Distinction*) - supervised by Prof. Mirco Musolesi.
Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning
Probabilistic and Unsupervised Learning¹ | Approximate Inference¹ | Bayesian Deep Learning
Applied ML | Supervised Learning | Reinforcement Learning² | Applied Deep Learning
- Imperial College London** October 2020 – June 2023
BSc Physics with Theoretical Physics (Honours)
BSc Thesis (*Distinction*) - supervised by Dr David Clements. *Computational Evolution*

Projects

- Transformer Program Synthesis — University College London** 2025
Developed novel programming-by-example framework: Transformer-based model recovers algorithms from data, training neural functions on-the-fly.
- Masked Image Modelling — University College London** 2024
Self-supervised learning framework: VisionTransformer pre-trained to reconstruct partially masked images and fine-tuned on Oxford-IIIT Pets dataset for segmentation. (*Distinction*)
- Machine learning for particle physics — Imperial College London** 2022
Lead ML team analysing decay product distribution from a Large Hadron Collider experiment (LHCb) for Standard Model comparison. TensorFlow, XGBoost. (*Distinction, highest score*)
- Solar Sail (Project Svarog) — Imperial College Space Society** 2022
Orbital trajectory simulation, atmospheric effects, relativistic corrections. Launching in 2028.

Skills & Languages

Python (PyTorch, TensorFlow, GPflow, JAX), Julia; GPU clusters, distributed training; Git, Linux
English, Hungarian, French (Intermediate)

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind